

2	A stationary electron is in a uniform field of force.		
	(a)	Describe the direction of the force on the electron relative to the direction of the field if the field is	
		(i)	a gravitational field,
			 [1]
			same direction of the field. 1
		(ii)	an electric field,
			 [1]
			opposite direction to the field. 1
		(iii)	a magnetic field.
			 [1]
			no force acts. 1
	(b)	Describe and explain the path the electron will take when it moves at right angles to each field.	
			
			
			
			
			
			
			
			
			
		 [5]	
		For gravitational field and electric field , the electron will move in a parabolic path , as the <u>net force (or acceleration) is always of constant direction and constant magnitude throughout the motion</u> despite the direction of the electron's velocity. 1 1	
		For magnetic field , when the electron moves at right angle to the magnetic field, it <u>experiences a magnetic force that is always acting perpendicular to the velocity</u> of the electron. 1	
		This <u>changes the direction of the velocity of the electron continuously, but not the magnitude of velocity.</u> 1	

		This <u>constant speed will keep the magnetic force and hence the centripetal force constant in magnitude</u> , causing the electron to move in a <u>circular path of constant radius</u> .	1
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2	(a)	(i)	Explain what is meant by an <i>ideal gas</i> .
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